SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4   **Total Marks:40**

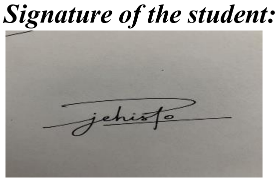
Annexure A

CASE STUDY

Code of Conduct:

Jehisto rijin J / 192511022 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu
Ii. If a patient has Fever AND Rash → Possible Measles
Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia
Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]
II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]
III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]
IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1.ANSWER: -

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The agent uses a Knowledge Base (KB) containing facts and rules. An inference mechanism is used to derive new information from the available facts.

(a) Propositional Logic Representation

Propositional Symbols

Let the following symbols represent the properties of Patient A:

- F → Patient A has Fever
- C → Patient A has Cough
- R → Patient A has Rash
- B → Patient A has Breathlessness
- Flu → Patient A has Possible Flu
- M → Patient A has Possible Measles
- P → Patient A has Possible Pneumonia Rules in the Knowledge Base

Rule I: Fever AND Cough → Possible Flu

$(F \land C) \rightarrow Flu$

Rule II: Fever AND Rash → Possible Measles

$(F \land R) \rightarrow M$

Rule III: Cough AND Breathlessness → Possible Pneumonia

$(C \land B) \rightarrow P$

Facts about Patient A

The given facts are:

$F=True$ $C=True$ $B=True$

Therefore, the initial knowledge base contains: $KB=\{F,C,B\}$

along with the three rules.

(b) Forward Chaining

Forward Chaining is a data-driven inference technique. It starts with the known facts and applies applicable rules to derive new facts.

Initial Facts

$F=True, C=True, B=True$

Inference 1 – Rule I

Rule I:

$(F \wedge C) \rightarrow Flu$

The conditions are:

- Fever = True
- Cough = True

Therefore:

$F \wedge C = True$

Hence Rule I is fired.

$Flu = True$

Derived fact: Patient A has Possible Flu.

Inference 2 – Rule II

Rule II:

$(F \wedge R) \rightarrow M$

Fever is available:

$F = True$

But there is no fact that Patient A has Rash.

Therefore:

$R = Unknown$

So the rule cannot be fired.

Possible Measles cannot be derived

Inference 3 – Rule III

Rule III:

$(C \wedge B) \rightarrow P$

The conditions are:

$C = True$ $B = True$

Therefore:

$C \wedge B$=True

Rule III is fired.

P=True

Derived fact: Patient A has Possible Pneumonia.

Forward Chaining Table

Step	Rule	Condition	Result
0	Initial Facts	F, C, B	Fever, Cough, Breathlessness
1	Rule I	$F \wedge C$ = True	Flu
2	Rule II	$F \wedge R$ = Unknown	Not fired
3	Rule III	$C \wedge B$ = True	Pneumonia

Forward Chaining Result

Flu=True Pneumonia=True

Possible Measles is not derived because Rash is not available as a fact.

(c) Backward Chaining

Backward Chaining is a goal-driven inference technique. It starts from the desired goal and works backward to determine whether the goal can be proved using the rules and known facts.

Given Goal

The goal is:

P=Pneumonia

We search the Knowledge Base for a rule whose conclusion is Pneumonia.

Rule III is:

$(C \wedge B) \rightarrow P$

Therefore, to prove Pneumonia, we must prove two sub-goals:

C=Cough and

B=Breathlessness

Step 1: Reduce the Goal

Goal:

P

Using Rule III:

$(C \land B) \rightarrow P$

The goal is reduced to:

$C \land B$

Step 2: Verify Cough

The knowledge base contains:

C=True

Therefore, the first sub-goal is satisfied.

Cough=True

Step 3: Verify Breathlessness

The knowledge base contains:

B=True

Therefore, the second sub-goal is satisfied.

Breathlessness=True

Step 4: Prove the Goal

Since both sub-goals are true:

$C \land B$=True

Therefore Rule III can be applied:

$(C \wedge B) \rightarrow P$

Hence:

$P = True$

Therefore, the goal is successfully proved.

Final Conclusion

The rule-based Logical Agent successfully uses propositional logic and inference techniques to diagnose Patient A.

Using Forward Chaining:

$F \wedge C \rightarrow Flu$

gives: Possible

Flu   and

$C \wedge B \rightarrow Pneumonia$

gives:

Possible Pneumonia

Using Backward Chaining, the goal Pneumonia is reduced to Cough AND Breathlessness. Since both are given as true facts, the goal is proved.

∴ Patient A has Possible Pneumonia

Thus, the Logical Agent confirms Possible Pneumonia based on the given rules and facts.

2. ANSWER: -

(a) Unification

Rule 1:

$\forall x Vehicle(x) \land EmergencyType(x) \rightarrow ClearPath(x)$ Given

facts:

Vehicle(Ambulance) EmergencyType(Ambulance)

Step 1:

Match:

Vehicle(x) with:

Vehicle(Ambulance) Therefore:

$\theta 1 = \{x/Ambulance\}$

Step 2:

Substitute x=Ambulance: EmergencyType(Ambulance)

This matches the given fact.

Therefore, the **MGU** is: $\theta = \{x/Ambulance\}$

Hence:

ClearPath(Ambulance)

(b) Resolution Proof

Goal:

Proceed(CarA)

CNF of Rule 1:

$\neg Vehicle(x) \lor \neg EmergencyType(x) \lor ClearPath(x)$

CNF of Rule 2:

$\neg ClearPath(x) \lor GreenSignal(x)$

CNF of Rule 3:

$\neg Vehicle(x) \lor \neg Behind(x,y) \lor \neg GreenSignal(y) \lor Proceed(x)$

Facts:

Vehicle(Ambulance) EmergencyType(Ambulance) Vehicle(CarA)

Behind(CarA,Ambulance)

Resolution Step 1

Using Rule 1:

Vehicle(Ambulance)+EmergencyType(Ambulance)

gives:

ClearPath(Ambulance)

Resolution Step 2

Using Rule 2:

ClearPath(Ambulance) gives:

GreenSignal(Ambulance)

Resolution Step 3

Rule 3 with:

x=CarA,y=Ambulance

and the facts:

Vehicle(CarA), Behind(CarA,Ambulance), GreenSignal(Ambulance) gives:

Proceed(CarA)

Resolution Chain

Vehicle(Ambulance)

EmergencyType(Ambulance)

ClearPath(Ambulance)

GreenSignal(Ambulance)

Vehicle(CarA)

Behind(CarA,Ambulance)

↓

Proceed(CarA)

Therefore:

Proceed(CarA) is TRUE

(c) Two Simultaneous Emergency Vehicles

The current KB can represent two emergency vehicles because the rules use **∀x**, so they can apply to different vehicles.

For example:

Vehicle(Amb1), EmergencyType(Amb1) Vehicle(Amb2), EmergencyType(Amb2)

The system can derive:

ClearPath(Amb1), ClearPath(Amb2) and

their green signals.

Problem

The current KB does **not** know how to handle conflicts if both vehicles need the same junction/path.

Additional rules/facts needed:

1. **Location:**
At(Amb1,Junction1) 2.
Path conflict:
PathConflict(Amb1,Amb2)
3. **Priority rule:**
Priority(Amb1,Amb2)
4. **Waiting rule:**
If paths conflict, one vehicle should wait until the other passes. **Conclusion**

KB supports multiple vehicles but needs priority and conflict rules for safe coordination.

3.ANSWER: -

(a) Convert into CNF

P1:

$SoilDry \rightarrow IrrigationNeeded$ Eliminate

implication:

$\neg SoilDry \lor IrrigationNeeded$

This is already in CNF.

P2:

$IrrigationNeeded \land CropWheat \rightarrow ApplyDripMethod$ Eliminate

implication:

$\neg(IrrigationNeeded \land CropWheat) \lor ApplyDripMethod$ Using

De Morgan's law:

$\neg IrrigationNeeded \lor \neg CropWheat \lor ApplyDripMethod$ Therefore:

$\neg IrrigationNeeded \lor \neg CropWheat \lor ApplyDripMethod$

P3:

$\neg ApplyDripMethod \land CropWheat \rightarrow CropAtRisk$

Eliminate implication:

$\neg(\neg ApplyDripMethod \land CropWheat) \lor CropAtRisk$ Using

De Morgan's law:

$ApplyDripMethod \lor \neg CropWheat \lor CropAtRisk$ Therefore:

$ApplyDripMethod \lor \neg CropWheat \lor CropAtRisk$

Facts

Given:

$SoilDry = True$ $CropWheat = True$

CNF:

SoilDry CropWheat

Final CNF

¬SoilDry∨IrrigationNeeded ¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod ApplyDripMethod∨¬CropWheat∨CropAtRisk SoilDry CropWheat

(b) Resolution Refutation

Goal:

ApplyDripMethod

For **Resolution Refutation**, negate the goal:

¬ApplyDripMethod

Add it to the KB.

Step 1: Derive IrrigationNeeded

From:

¬SoilDry∨IrrigationNeeded and:

SoilDry

Resolution gives:

IrrigationNeeded

Step 2: Derive ApplyDripMethod

Use:

¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod

We have:

IrrigationNeeded

and: CropWheat

Therefore:

ApplyDripMethod

Step 3: Contradiction

We added the negated goal:

¬ApplyDripMethod But

we derived:

ApplyDripMethod

Therefore:

ApplyDripMethod∨¬ApplyDripMethod

Resolution gives:

□

The empty clause means a contradiction has been obtained.

Resolution Proof Tree

SoilDry
+
¬SoilDry ∨ IrrigationNeeded
↓
IrrigationNeeded
+
CropWheat
+
¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
↓
ApplyDripMethod
+
¬ApplyDripMethod ← Negated Goal
↓
□

Therefore:

ApplyDripMethod is TRUE

(c) If SoilDry is Removed

Initially:

SoilDry=True was

used to derive:

IrrigationNeeded

Then:

IrrigationNeeded+CropWheat gave:

ApplyDripMethod

After removing SoilDry:

We can no longer apply P1: SoilDry→IrrigationNeeded

Therefore:

IrrigationNeeded cannot be derived

Since IrrigationNeeded is unavailable, P2 cannot be applied:

IrrigationNeeded∧CropWheat→ApplyDripMethod

Therefore:

ApplyDripMethod cannot be concluded

What about CropAtRisk?

P3 is:

¬ApplyDripMethod∧CropWheat→CropAtRisk

But we **cannot assume**: ¬ApplyDripMethod just

because ApplyDripMethod cannot be proved.

This is called the **Closed World fallacy / failure to prove is not proof of negation**.

Therefore:

CropAtRisk cannot be concluded

Final Result

Situation	Conclusion
SoilDry present	IrrigationNeeded ✓

SoilDry present	ApplyDripMethod ✓
SoilDry removed	IrrigationNeeded cannot be derived
SoilDry removed	ApplyDripMethod cannot be derived
SoilDry removed	**CropAtRisk cannot be concluded**

Removing SoilDry breaks the inference chain, but does not prove CropAtRisk.

4.ANSWER :-

(a) Belief State and Modus Ponens

Given Percepts

The agent perceives: Stench[1,2]

Breeze[1,1] Glitter[2,2] Therefore,

the initial belief state is:

{Stench[1,2], Breeze[1,1], Glitter[2,2]}

Apply Modus Ponens

Rule 1:

Stench[1,2]→WumpusAdjacent[1,2]

Since:

Stench[1,2]=True we

derive:

WumpusAdjacent[1,2]

Rule 2:

Breeze[1,1]→PitAdjacent[1,1]

Since:

Breeze[1,1]=True we

derive:

PitAdjacent[1,1]

Rule 3:

Glitter[x,y]→Gold[x,y] For

[x,y]=[2,2]:

Glitter[2,2]→Gold[2,2]

Therefore:

Gold[2,2]

Derived Knowledge

WumpusAdjacent[1,2], PitAdjacent[1,1], Gold[2,2]

(b) Forward Chaining – Possible Locations

In a standard Wumpus grid, adjacent cells to a position are horizontal and vertical neighbors.

From Stench [1,2]

A Wumpus can be in a cell adjacent to [1,2]:

[1,1], [1,3], [2,2]

So possible Wumpus locations are:

Wumpus∈{[1,1],[1,3],[2,2]}

From Breeze [1,1]

The cells adjacent to [1,1] are:

[1,2], [2,1]

Therefore, possible Pit locations are:

Pit∈{[1,2],[2,1]}

From Glitter [2,2] Glitter[2,2]→Gold[2,2]

Therefore:

Gold at [2,2]

Forward Chaining Summary

Stench[1,2]
↓
Possible Wumpus:

[1,1], [1,3], [2,2]

Breeze[1,1]
↓
Possible Pit:

[1,2], [2,1]

Glitter[2,2]
↓
Gold[2,2]

(c) Is the Path Safe?

Path:

[1,1]→[1,2]→[2,2]

We need to check each cell.

Cell [1,1]

From Breeze [1,1], there may be a pit in an adjacent cell.

But there is no direct evidence that [1,1] itself contains a Pit or Wumpus.

So:
[1,1] is not proven dangerous

Cell [1,2]

From the Breeze at [1,1]:
$Pit \in \{[1,2],[2,1]\}$
Therefore, [1,2] could contain a Pit.

Also, Stench at [1,2] indicates a Wumpus is adjacent to [1,2].

Thus [1,2] is not confirmed safe.

[1,2] is potentially dangerous

Cell [2,2]

Glitter tells us:

Gold[2,2]

However, from Stench [1,2]:

$Wumpus \in \{[1,1],[1,3],[2,2]\}$

Therefore, [2,2] could contain the Wumpus.

So [2,2] is also not confirmed safe.

Final Conclusion

The path:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is NOT

guaranteed to be safe.

Reason:

- [1,2] may contain a Pit.

- [2,2] may contain the Wumpus.
- The agent does not have enough information to prove these cells are safe.

Therefore:

The agent should NOT assume this path is safe.

Important: In Wumpus World, “not proven dangerous” is not the same as “proven safe.”